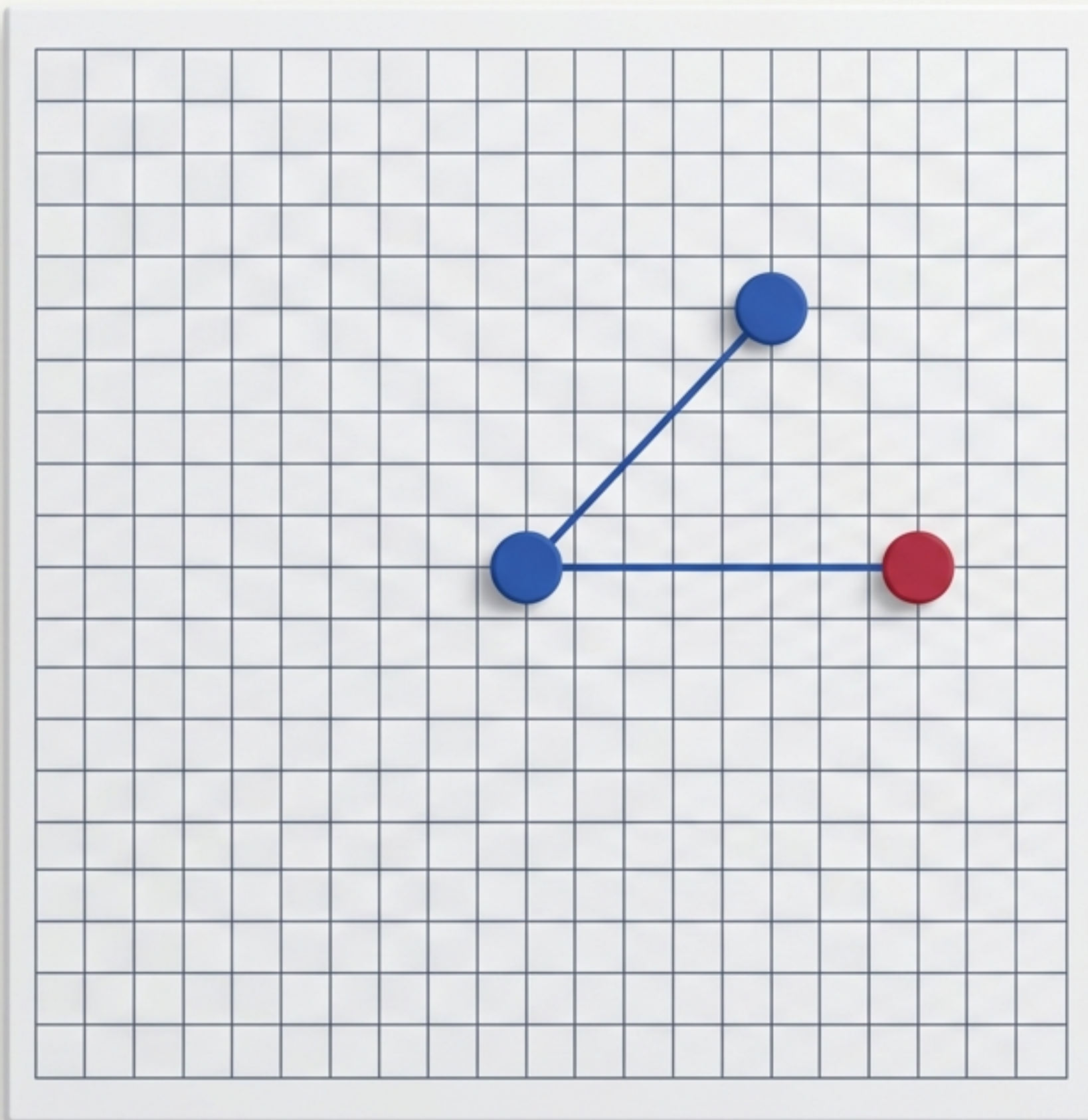
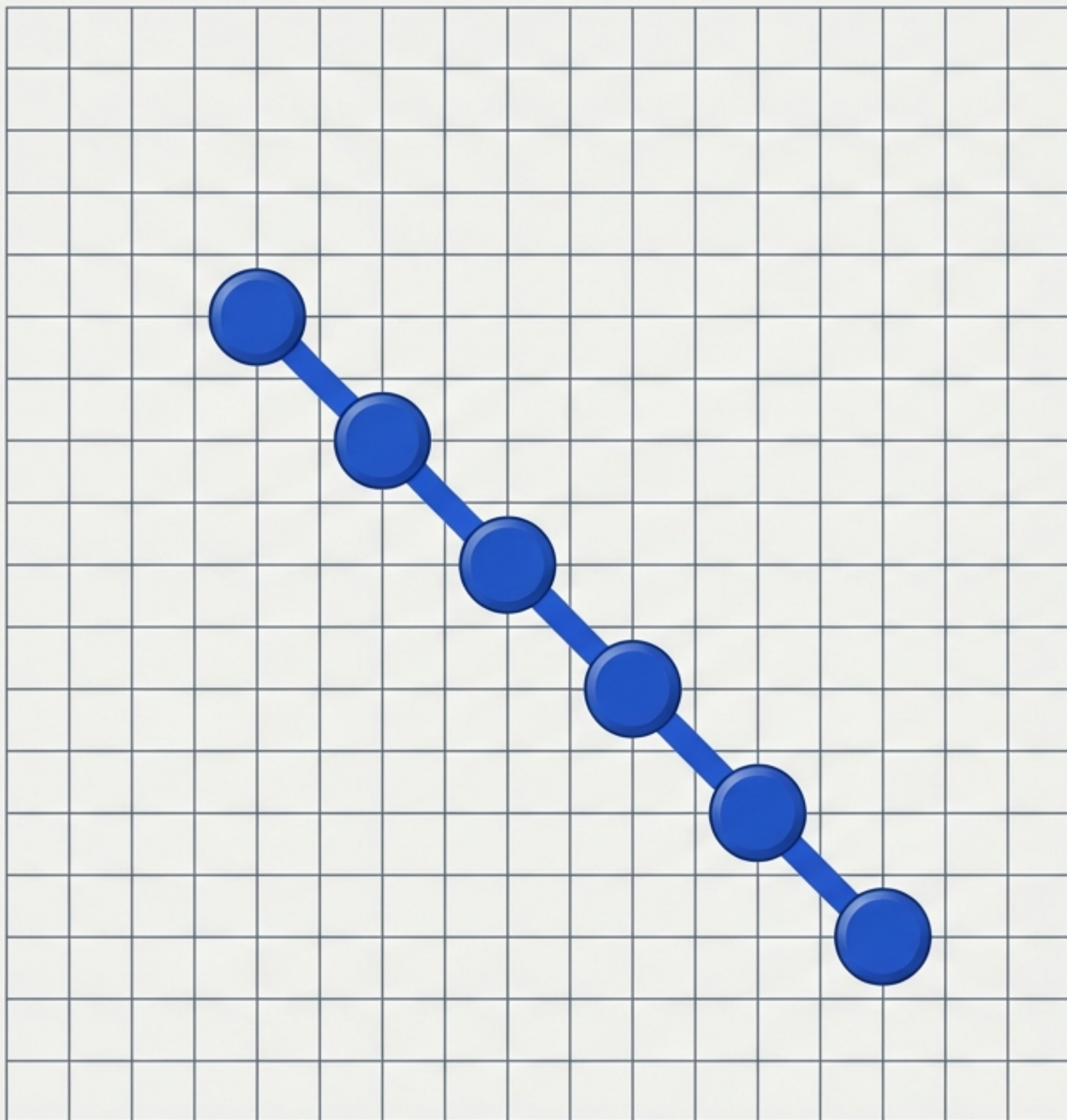




GOMOKU: The AI Engine

Yeah, well, your brain has to fry sometime.



The Prime Directive

Build an uncrashable AI that consistently beats human players on a standard 19x19 Goban, with no limit on the number of stones.

The baseline win condition is aligning 5 or more stones.

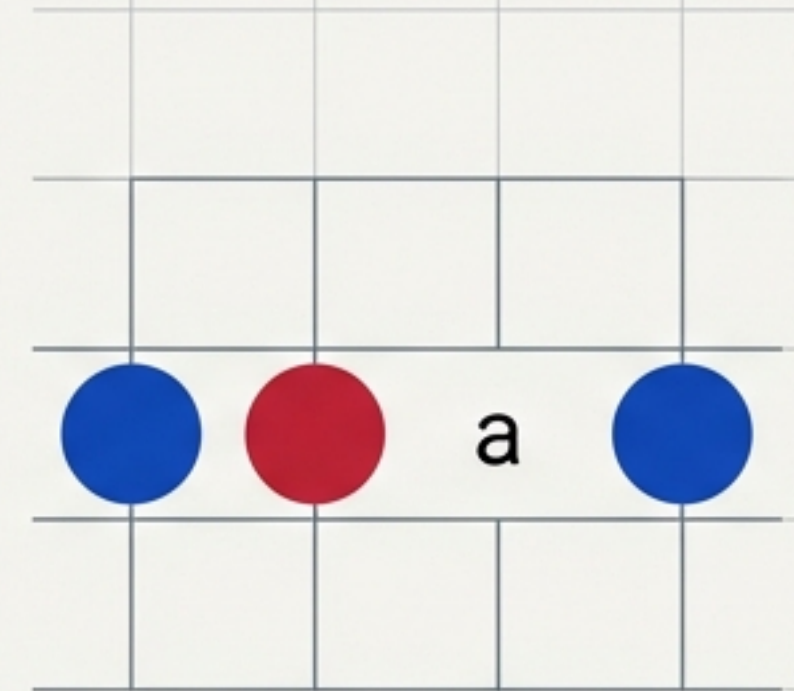
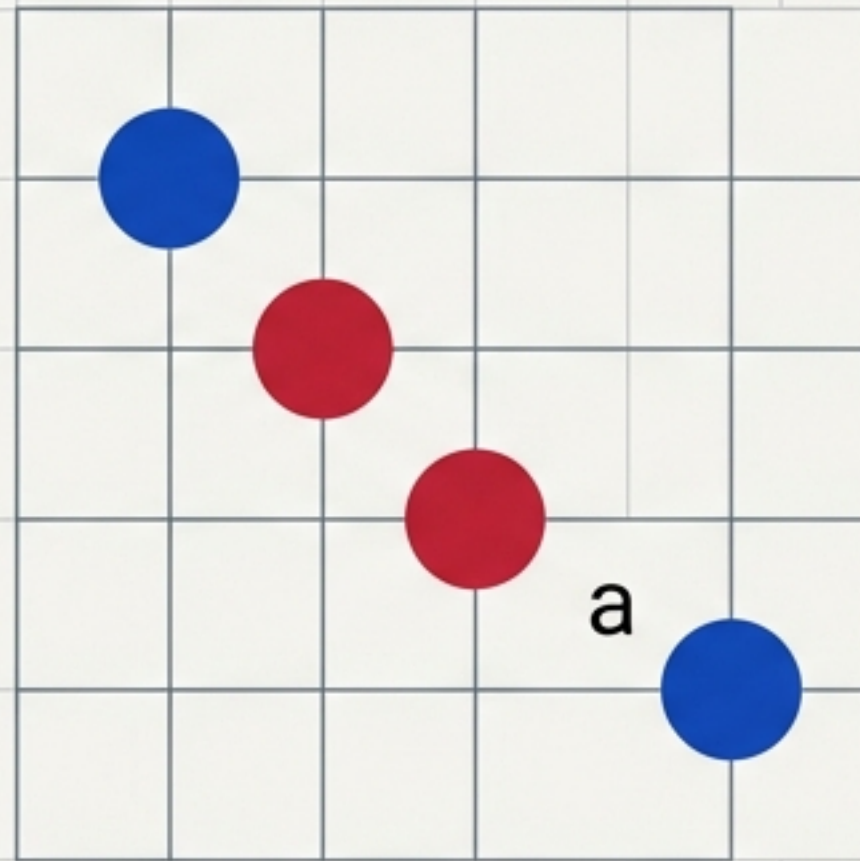


ZERO TOLERANCE: The program must not crash under any circumstances (including out-of-memory) or quit unexpectedly.
Penalty: Instant Grade 0.

Offensive Maneuvers: The Capture Rule

You can remove exactly a pair of opponent stones by flanking them. The newly freed intersections can be played on immediately.

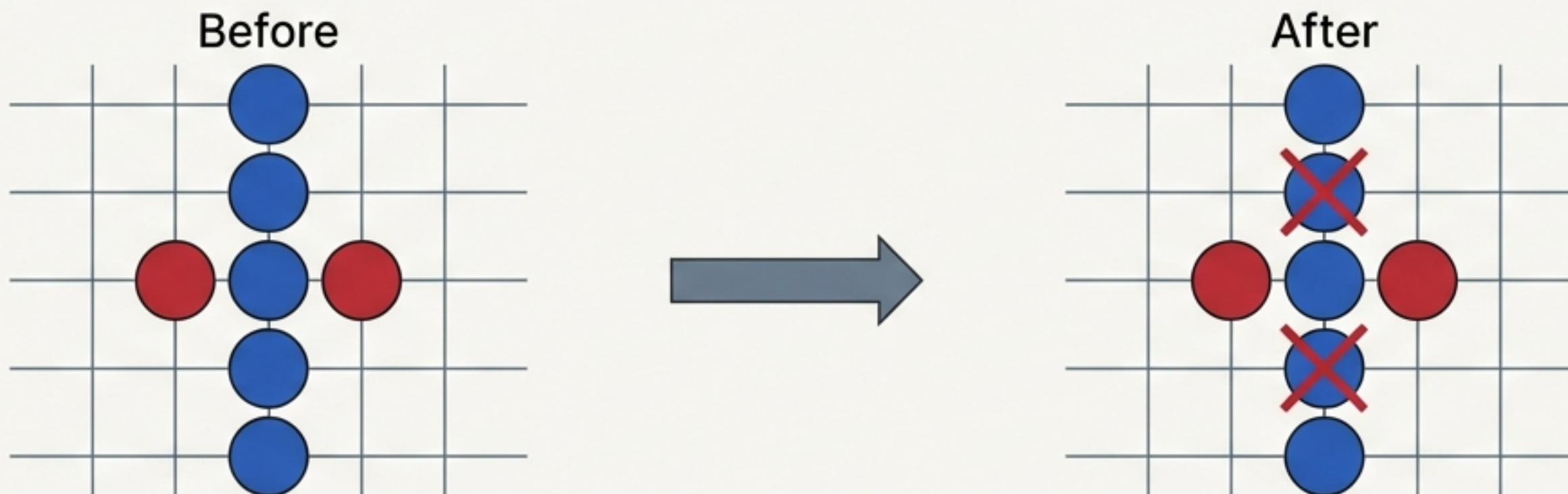
INSTANT WIN: Capturing 10 opponent stones (5 pairs) results in an immediate victory.



Vulnerable Victories: Endgame Captures

A 5-stone alignment does not guarantee a win if the opponent can break the line via capture.

Critical Exception: If a player has already lost 4 pairs, and the opponent captures a 5th pair while breaking the 5-alignment, the opponent wins by capture instead.

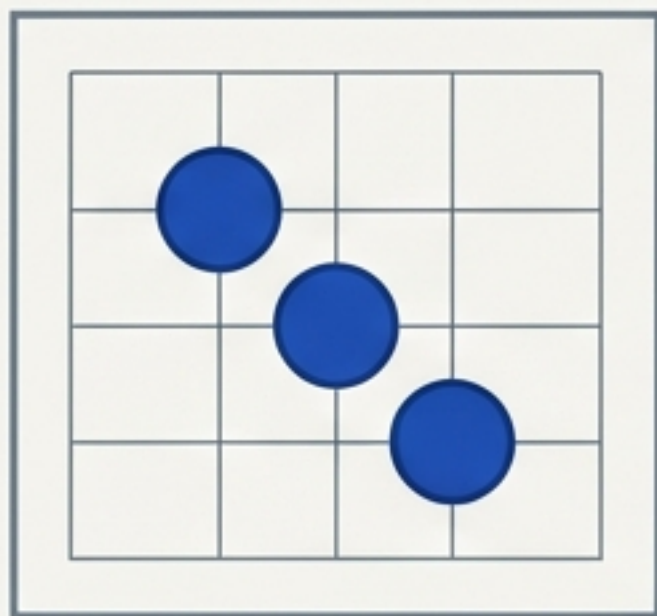


Maintaining Balance: No Double-Threes

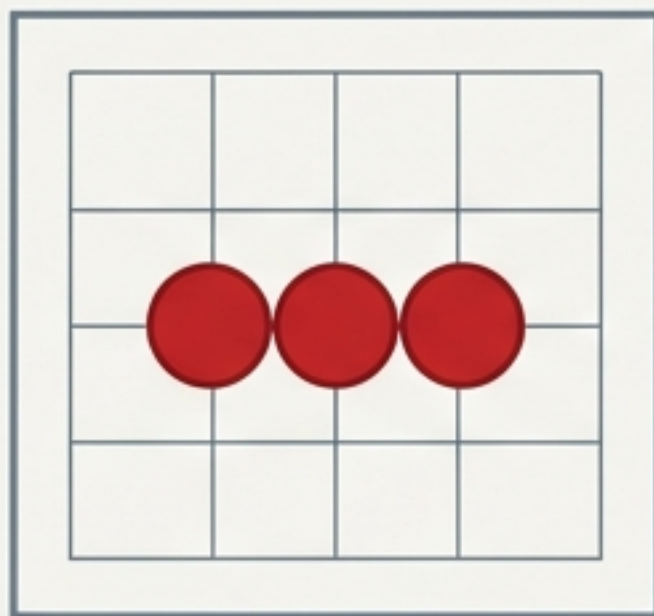
A free-three is a 3-stone alignment that, if unblocked, creates an indefensible 4-stone alignment.

It is strictly forbidden to play a single move that introduces two simultaneous free-three alignments.

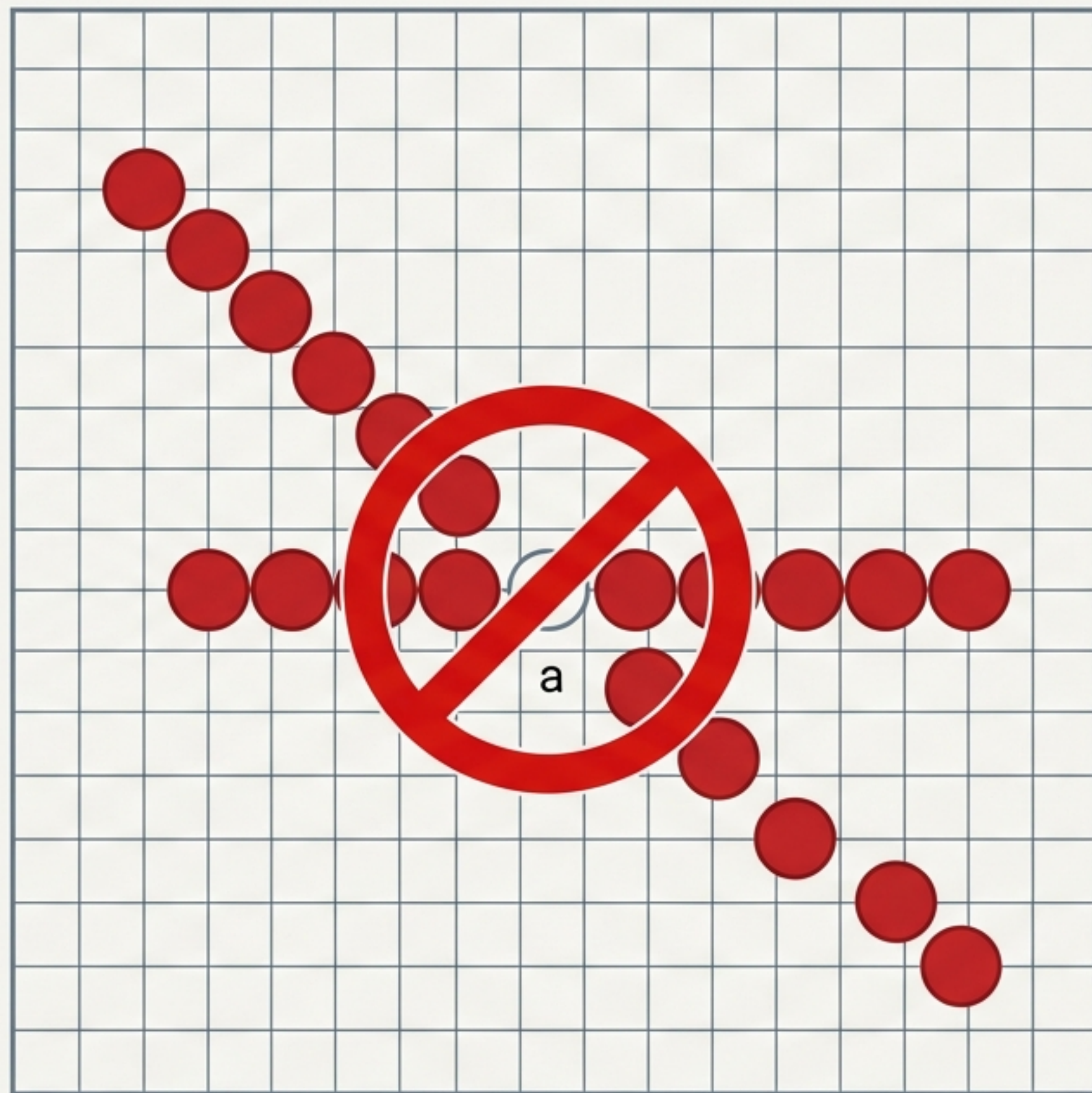
Note: Introducing a double-three by capturing a pair is perfectly legal.



Diagonal Free-Three



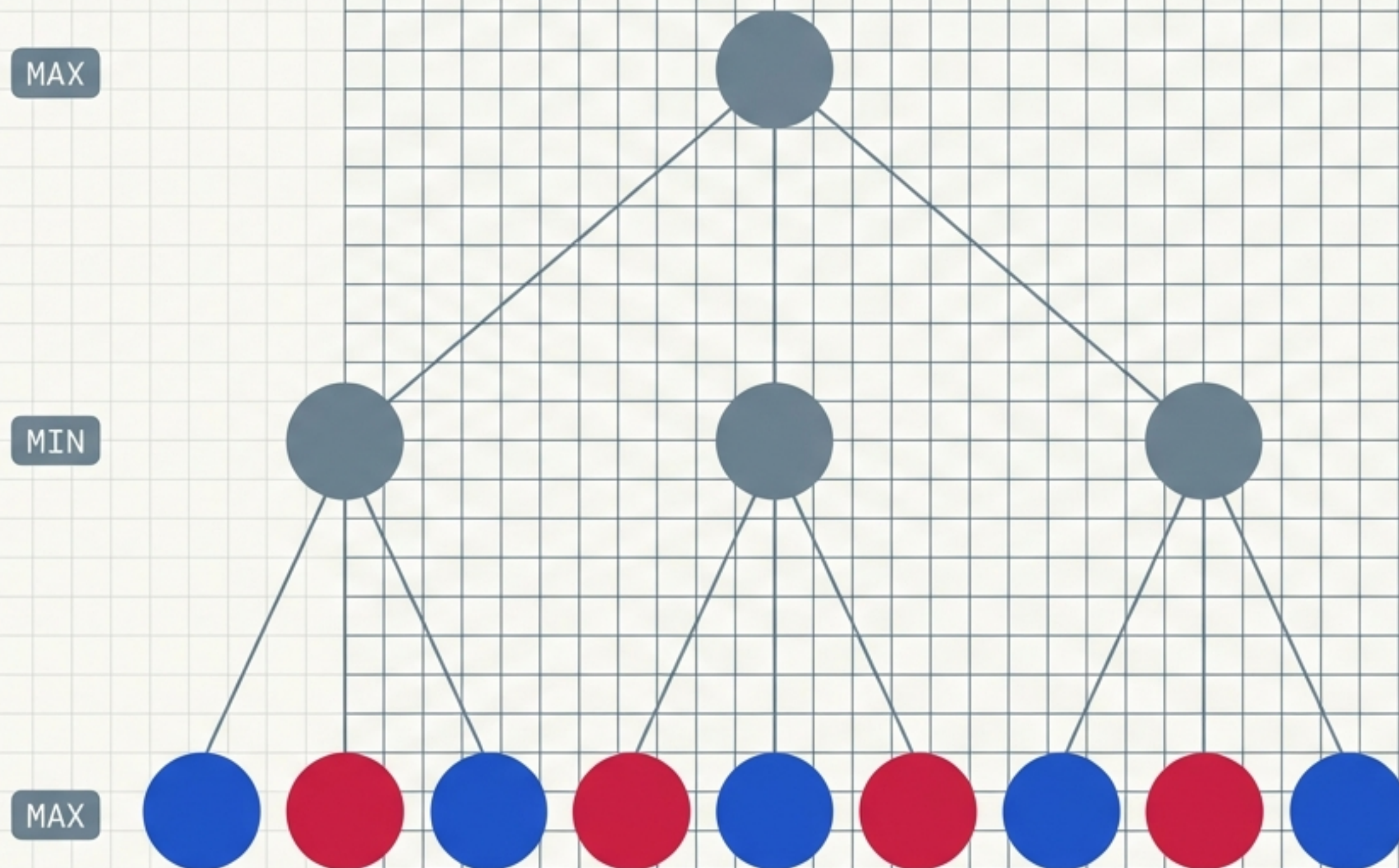
Horizontal Free-Three



The Engine Architecture: Minimax

The AI must generate a possible-solution tree to predict and counter human strategies. You are required to implement a Minimax algorithm (or variant).

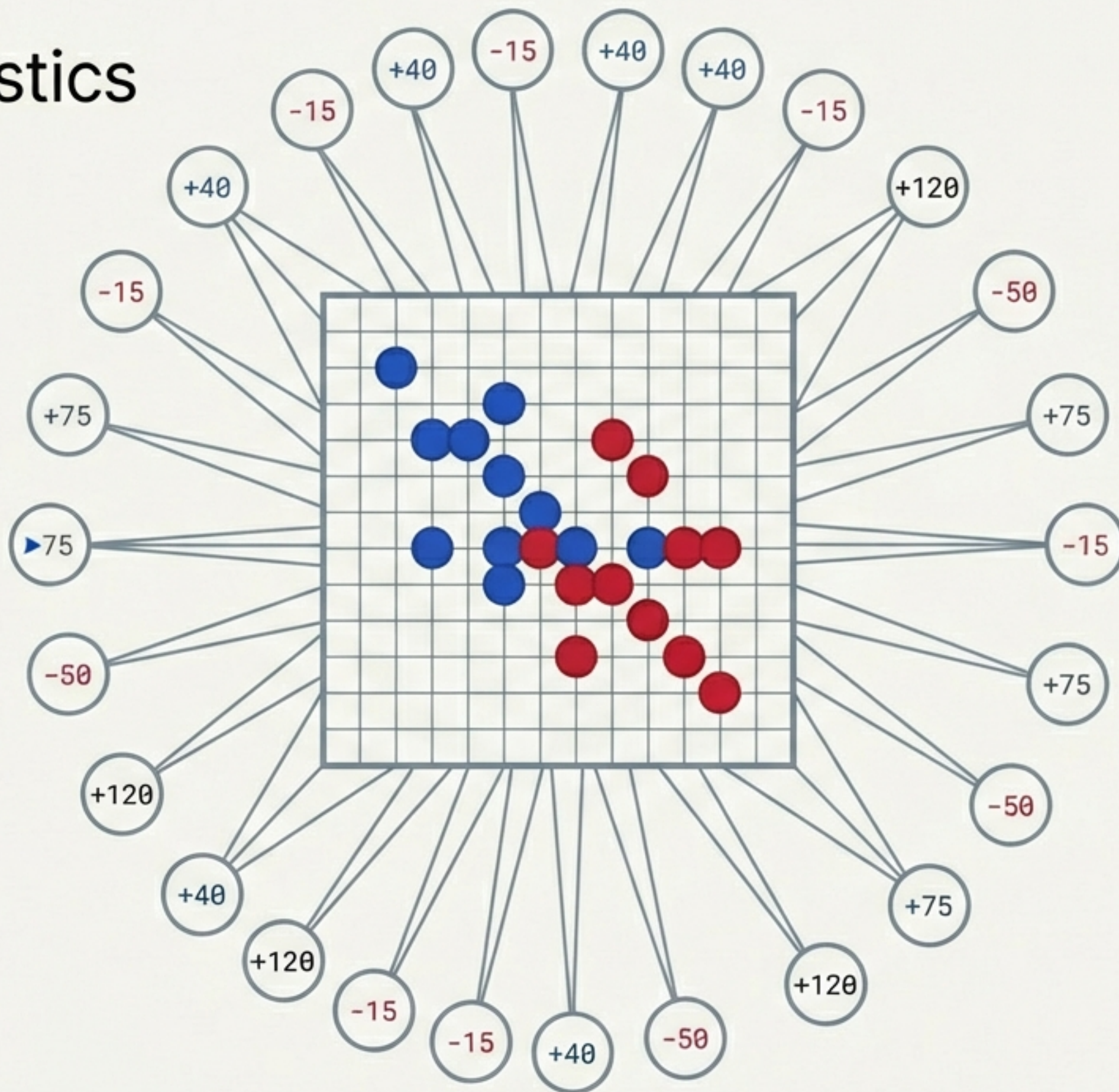
The algorithms themselves are well-documented. The implementation is just the foundation; the true difficulty lies in evaluation.



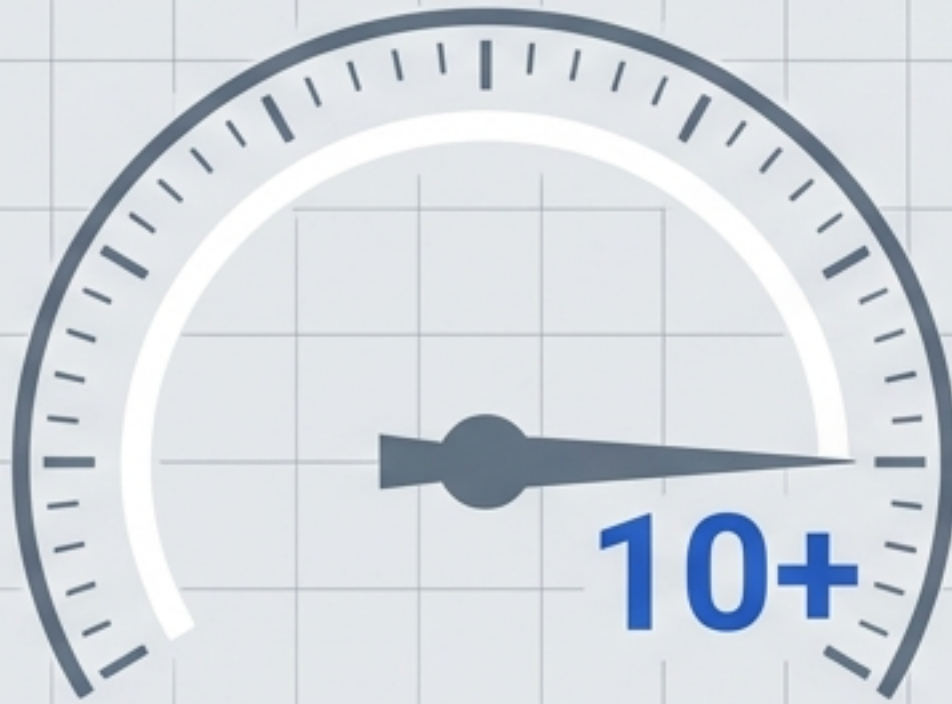
The True Challenge: Heuristics

Minimax is useless without an efficient heuristic function to evaluate the value of terminal nodes. It must balance extreme accuracy with blazing speed.

Don't be fooled: The heuristic is actually the hardest part.



Mandatory Performance Thresholds



Tree Depth

Must search at least 10 levels deep. (Lower depth = no maximum grade, regardless of win rate).



Calculation Speed

Max 0.5 seconds average to find a move. (Slower = project validation fails).



Build Quality

Avoid lazy implementations. Efficiency is strictly evaluated.

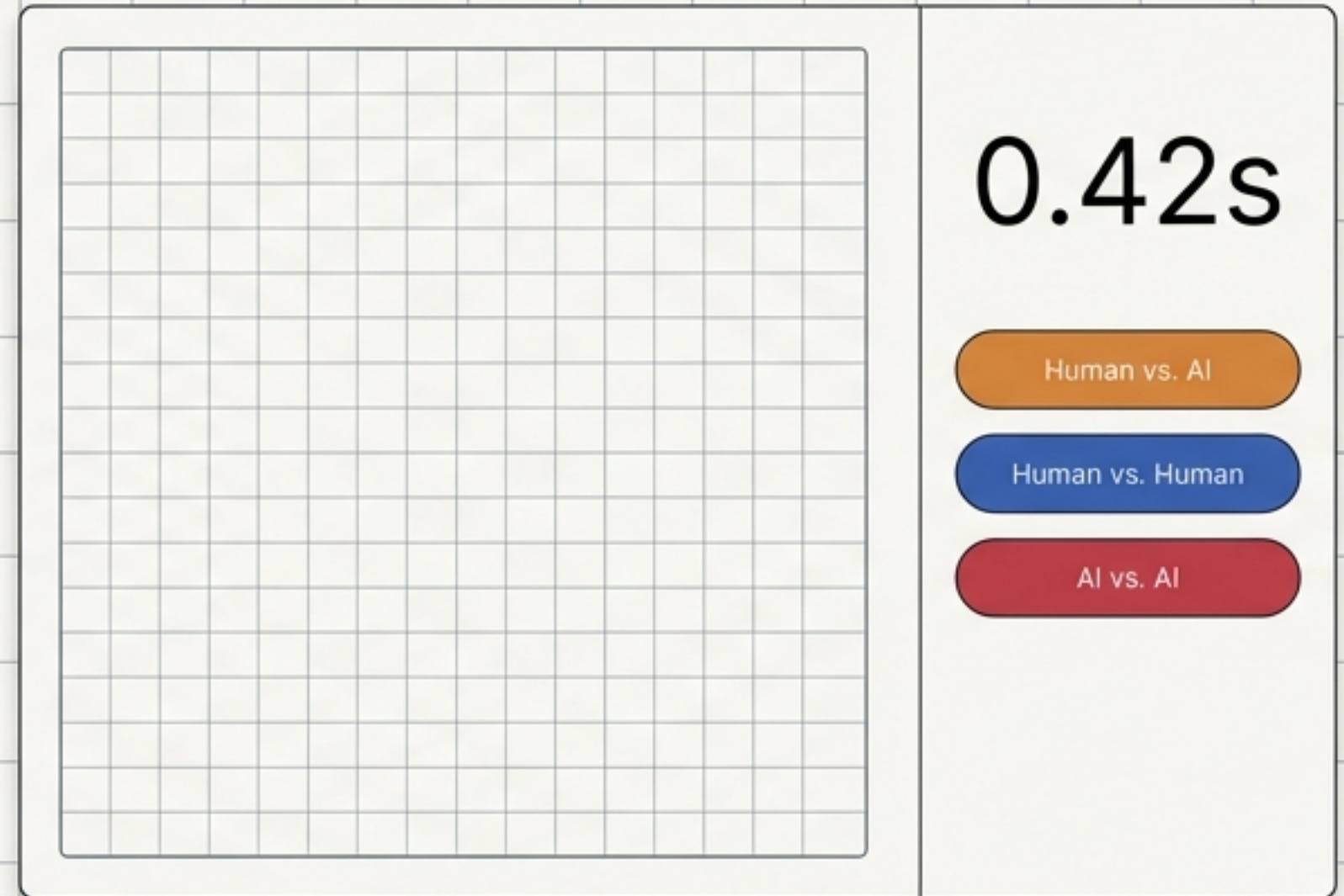
The Battleground Interface

Build a fully playable, visually pleasing GUI using any language or graphical/ncurses library.

Required Modes:

- Human vs. AI (AI must adapt, play to win, and not be let to win).
- Human vs. Human (Hotseat) featuring a live AI move-suggestion tool.

Required Deliverable: A Makefile producing the Gomoku executable with standard rules (all, clean, fclean, re). No relinking.



The Unforgiving Clock

The user interface must display a live timer counting exactly how long the AI takes to compute its next move. This is how the 0.5-second constraint constraint is verified.

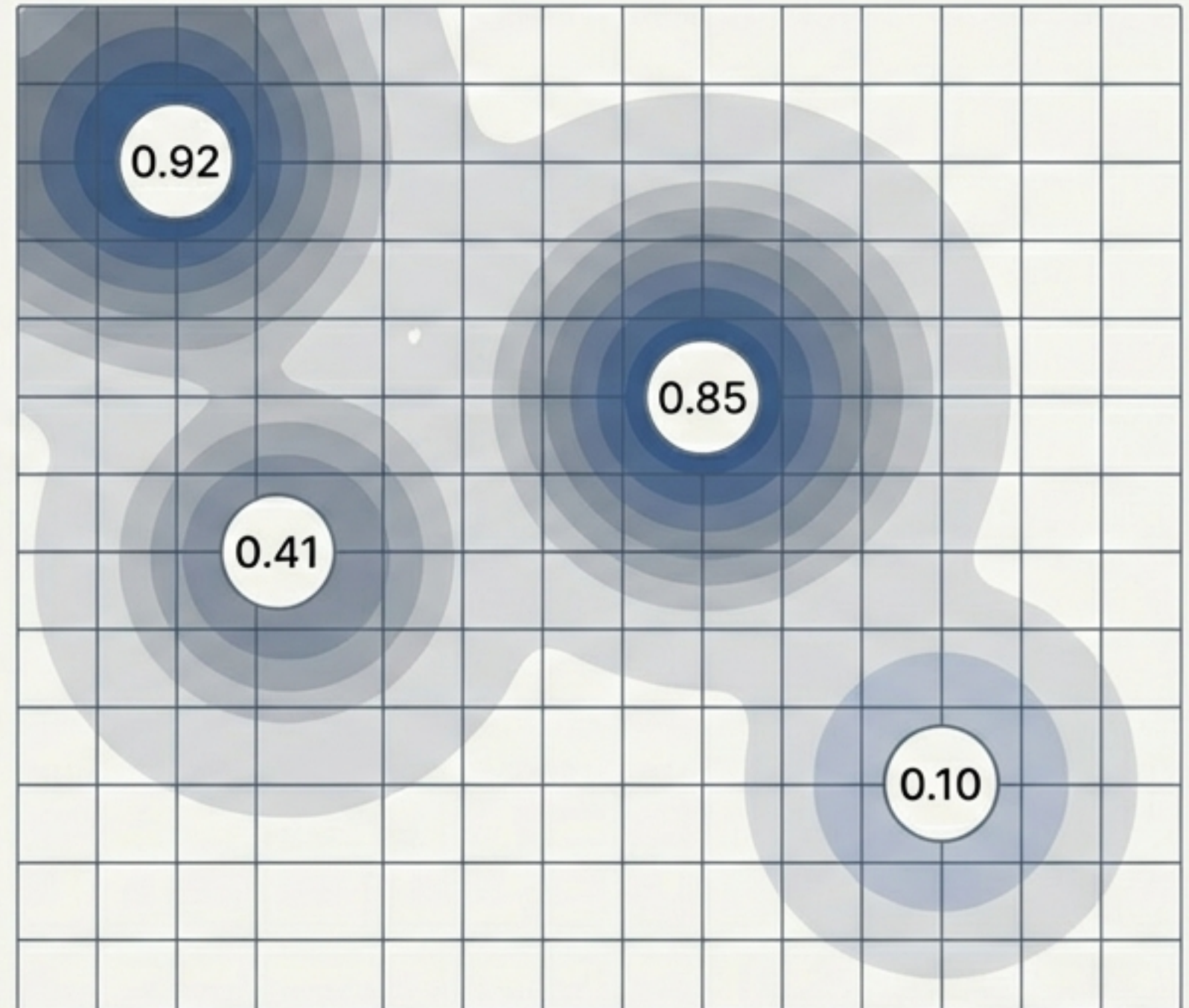
**No timer, no project validation.
It's that serious.**



Pro-Tip: Visualize the Brain

Highly recommended to build an interactive debugging mode.

Being able to see why the AI chose a move will refine your tactics and provide crucial visual proof during peer evaluation.



Surviving the Defense



Explain the Algorithm:

Detail your Minimax variant. If you can't explain it perfectly, you get zero points.



Defend the Heuristic:

Prove why it is accurate and fast.



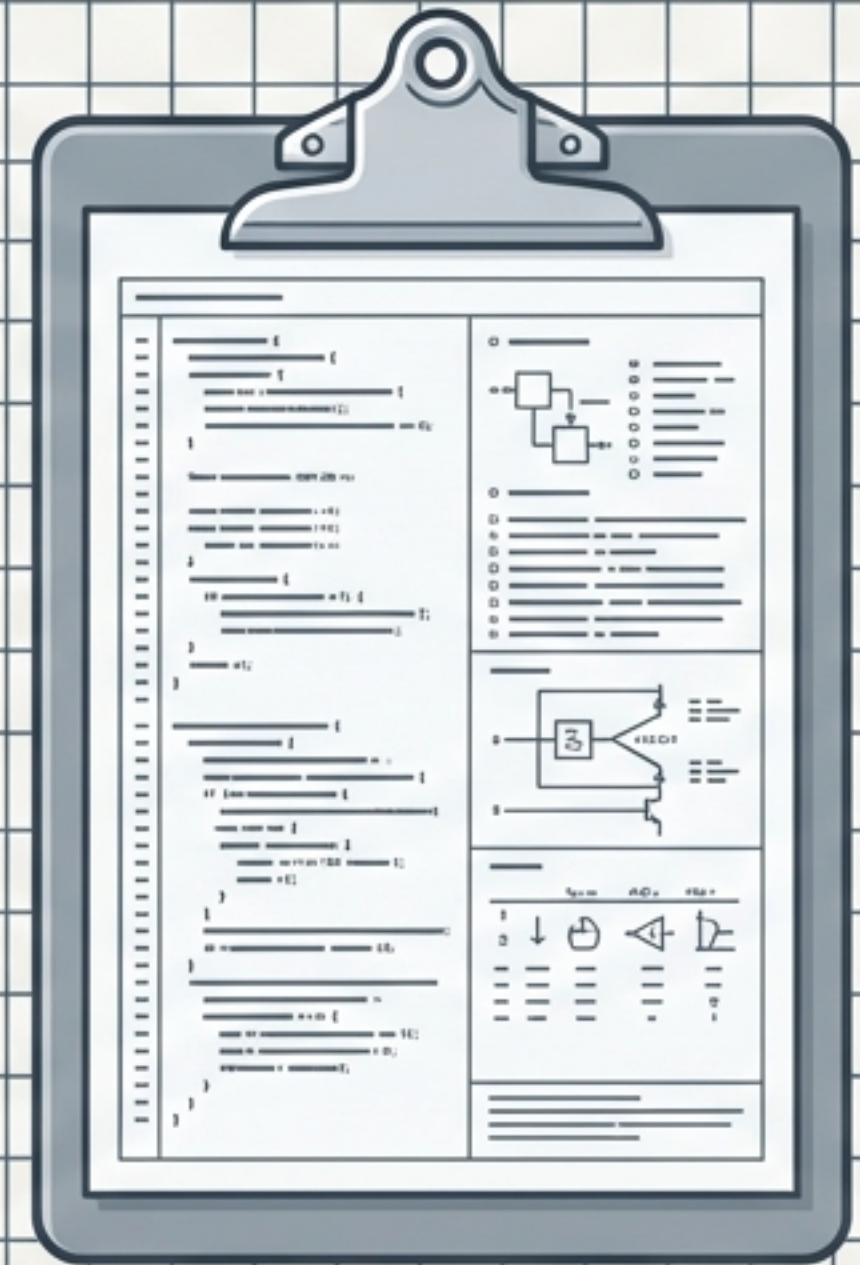
Prove the Rules:

Demonstrate correct implementation of captures, double-threes, and endgame exceptions.



Win the Duel:

Run the program against your grader...
and hope your AI doesn't get wiped out.



“For the love of Tartiflette, I can not stress the ‘thoroughly’ part enough.”

The Bonus Round: Perfection Required

Bonuses are strictly assessed only if the mandatory project is flawless—fully integrated and entirely bug-free.

Potential Upgrades: Allow users to select starting conditions (Standard, Pro, Swap, Swap2). Up to 5 custom, highly-polished bonuses will be evaluated. Make it interesting.

